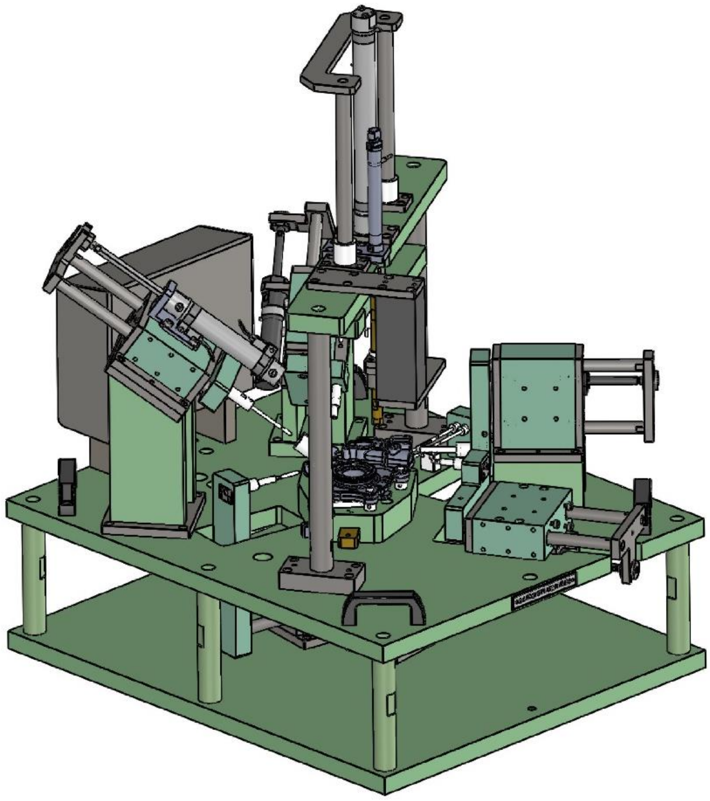
VAMTEC
MACHINES AND AUTOMATION PVT LTD

OPERATION AND MAINTENANCE MANUAL 2.6/2.7 FIXTURE



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MINDA CORPORATION LTD

DOCUMENT INFORMATION

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1.GENERAL INFORMATION

- This manual provides comprehensive guidelines for the operation, maintenance, and troubleshooting of ECM (Electro chemical Machining) fixtures.
- It is essential for operators and maintenance personnel to thoroughly understand and follow the instructions in this manual to ensure safe and efficient operation of the fixtures.

2.GENERAL SAFETY WARNINGS

- Personal Protective Equipment (PPE): Always wear appropriate PPE, including safety goggles, gloves, and protective clothing, when operating ECM fixtures.
- Electrical Safety: Ensure that the power supply is turned off before performing any maintenance. Use insulated tools and follow lockout/tag-out procedures.
- Chemical Handling: Handle electrolytes and other chemicals with care. Follow the Material Safety Data Sheets (MSDS) for proper storage, handling, and disposal.
- Emergency Procedures: Familiarize yourself with emergency shut-off procedures, first aid, and fire safety protocols.

3.OVERVIEW OF ECM FIXTURES

- ECM fixtures are designed to hold and position workpieces during the electrochemical machining process.
- They play a critical role in ensuring precision and consistency in machining operations.
- The fixtures are typically made from non-corrosive materials and are equipped with features to facilitate easy workpiece mounting, alignment, and removal.

NOTE:

- For 2.6 Fixture Top Electrode 2_2.6 Fix (VAM-P-463-02-02-00). This electrode needs to change for 2.6 Model part.
- For 2.7 Fixture Top Electrode 2_2.7 Fix (VAM-P-463-02-15-00). This electrode needs to change for 2.6 Model part.

4. GENERAL OPERATION GUIDELINES

4.1. Pre-Operation Checklist

- Inspect the ECM fixture for any visible damage or wear.
- Ensure that all connections including electrical, pneumatic and electrolyte lines for any visible damaged are view.

- Verify that the workpiece is correctly positioned and securely clamped in the fixture.
- Check that the electrolyte flow system is functioning correctly and that the electrolyte level is sufficient.
- Ensure that the power supply and control systems are functioning properly.

4.2. Deburring Process

4.2.1 Cycle Start-Up

- Turn on the ECM machine and initialize the fixtures.
- Select the appropriate deburring parameters (Voltage, Current, Electrolyte Flow, Deburring Time) as specified for the workpiece.
- Begin the Deburring process by starting the auto cycle.

4.2.2 Deburring

- Monitor the deburring process closely, ensuring that the electrolyte flow is consistent and that there are no blockages.
- Observe the fixture and workpiece for any signs of abnormal wear, vibration, or overheating.
- Make adjustments to the process parameters if necessary to maintain the precise deburring.

4.2.3 Cycle End

- Upon completion of the deburring process, the autocycle end.
- Carefully remove the workpiece from the fixture, ensuring that no damage occurs during removal.
- Wash with DM water after cycle complete.

5. MAINTENANCE PROCEDURES

5.1 Regular Maintenance

5.1.1 Daily

- Clean the fixture and surrounding area with DM water to remove any residual electrolyte or debris.
- Inspect the fixture for signs of corrosion, wear, or damage. Replace any worn or damaged components immediately.

5.1.2 Weekly

- Check the alignment and condition of clamping mechanisms. Lubricate moving parts as required.
- Inspect electrical connections for signs of wear or corrosion. Tighten or replace connectors as needed.

5.1.3 Monthly

- Perform a thorough inspection of the entire fixture, including all electrical and mechanical components.
- Test the electrolyte flow system for proper operation. Replace filters and clean lines as necessary.

5.2 Preventive Maintenance

- Refer the Preventive maintenance check list given by Vamtec Machine.
- Schedule periodic checks of the ECM machine and fixtures by qualified personnel.
- Replace worn components according to the manufacturer's recommendations.
- Keep a detailed maintenance log to track the condition of the fixture and any repairs or replacements performed.

6. TROUBLESHOOTING

6.1 Fixture Not Holding Workpiece

- Check for wear or damage to clamping mechanisms.
- Ensure the workpiece is properly aligned and seated in the fixture.

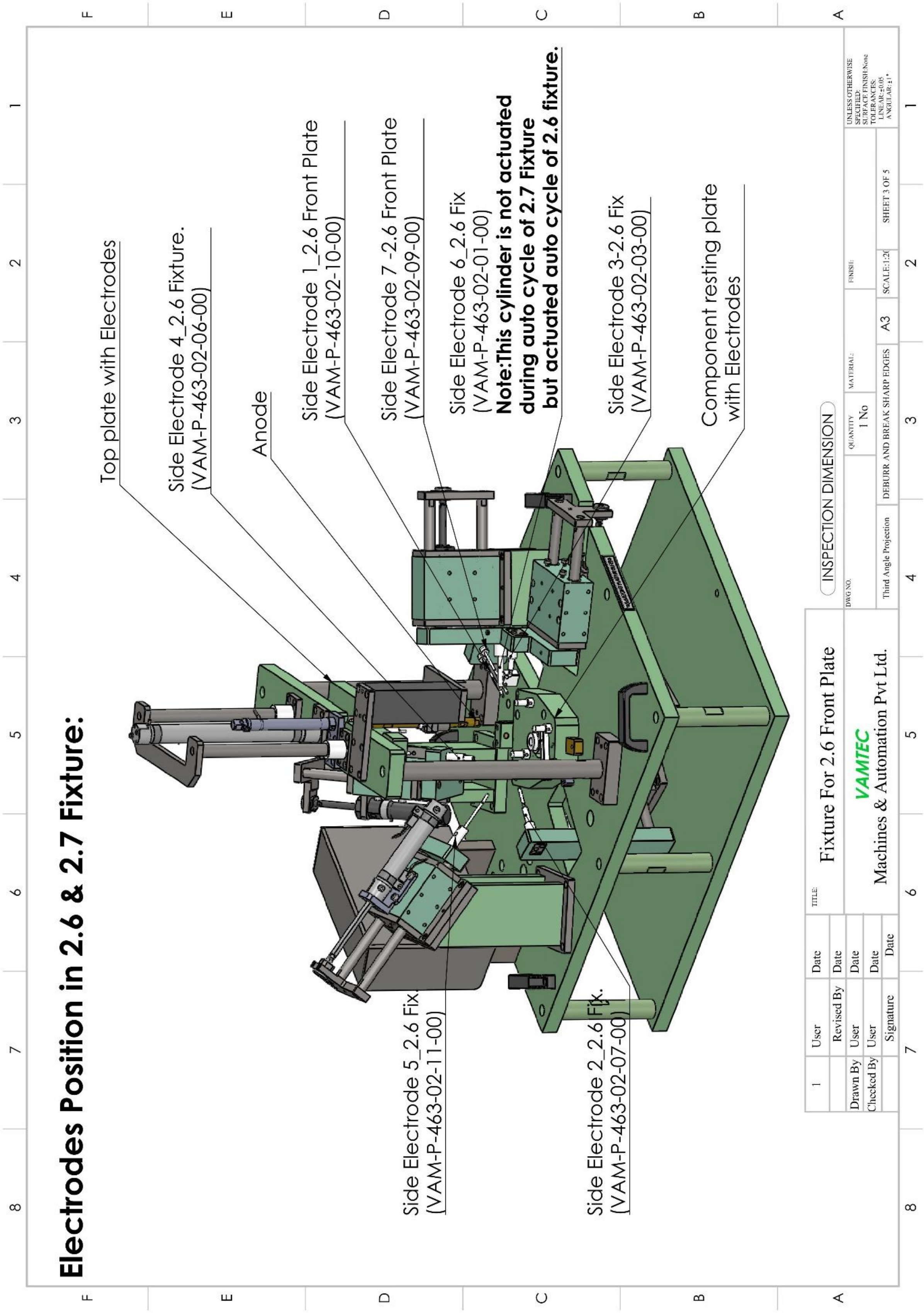
6.2 Electrolyte Flow Issues

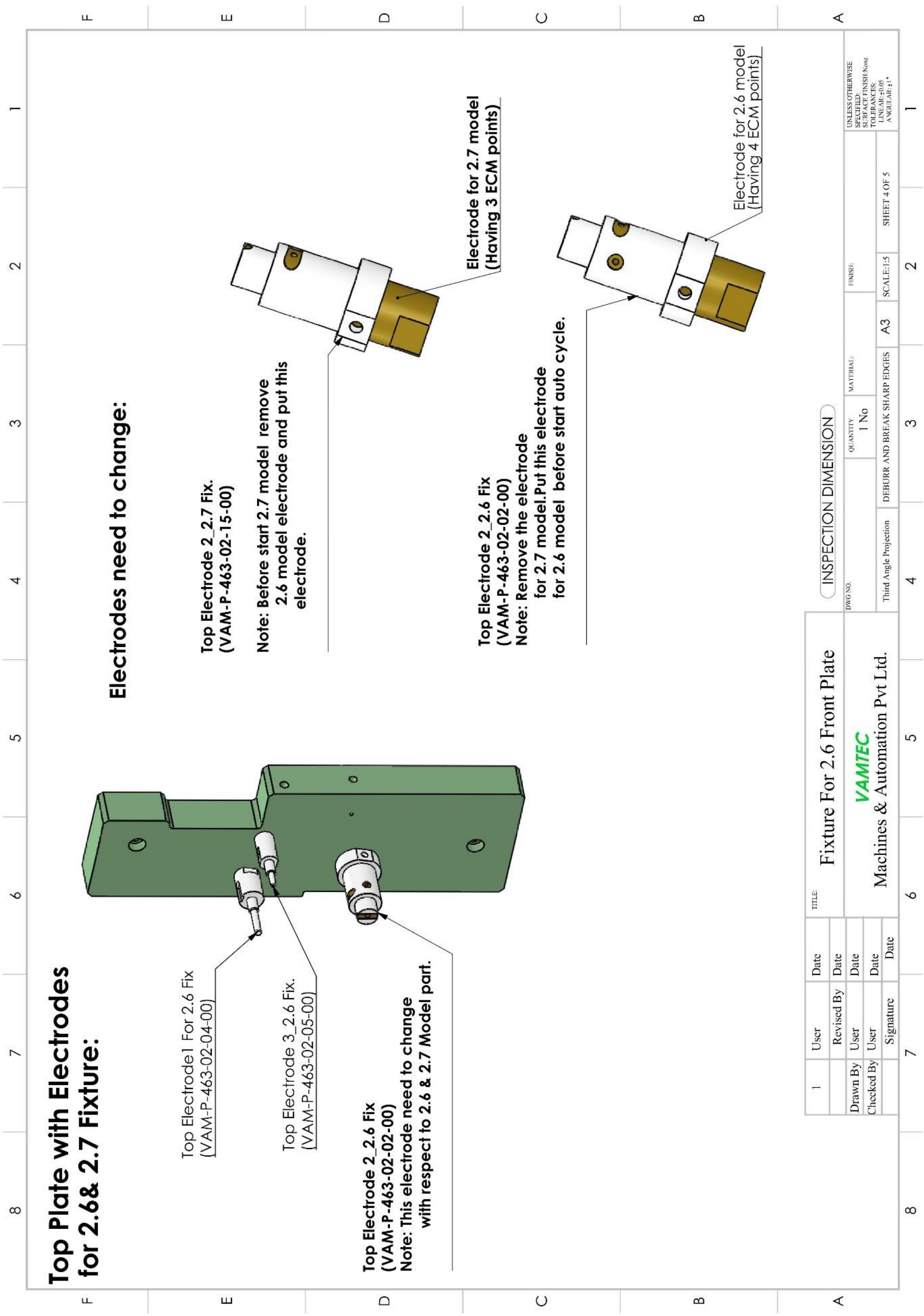
- Inspect for blockages in the electrolyte lines and electrolyte fittings.
- Check the electrolyte pump operation.

6.3 Short Circuit Issues

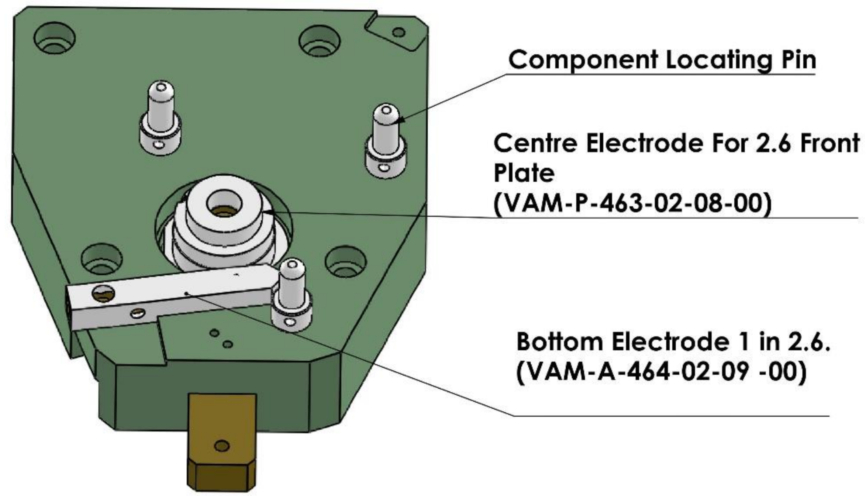
- Check the burr in the workpiece.
- Check the alignment of the workpiece and electrodes

7. FIXTURE DRAWINGS




Top Plate with Electrodes for 2.6& 2.7 Fixture:
Top Electrode1 For 2.6 Fix (VAM-P-463-02-04-00)
Top Electrode 3_2.6 Fix. (VAM-P-463-02-05-00)
Top Electrode 2_2.6 Fix
(VAM-P-463-02-02-00)
Note: This electrode need to change with respect to 2.6 & 2.7 Model part.
Electrodes need to change:
Top Electrode 2_2.7 Fix.
(VAM-P-463-02-15-00)
Note: Before start 2.7 model remove 2.6 model electrode and put this electrode.
Electrode for 2.7 model (Having 3 ECM points)
Top Electrode 2_2.6 Fix
(VAM-P-463-02-02-00)
Note: Remove the electrode for 2.7 model.Put this electrode for 2.6 model before start auto cycle.
Electrode for 2.6 model (Having 4 ECM points)
INSPECTION DIMENSION
Fixture For 2.6 Front Plate
VAMTEC
Machines & Automation Pvt Ltd.
Drawn By User
Checked By User
Revised By
Signature
Date
QUANTITY 1 No
DEBURR AND BREAK SHARP EDGES
Third Angle Projection
A3
SCALE:1:5
SHEET 4 OF 5

Component Resting plate with Electrodes for 2.6 & 2.7 Fixture:

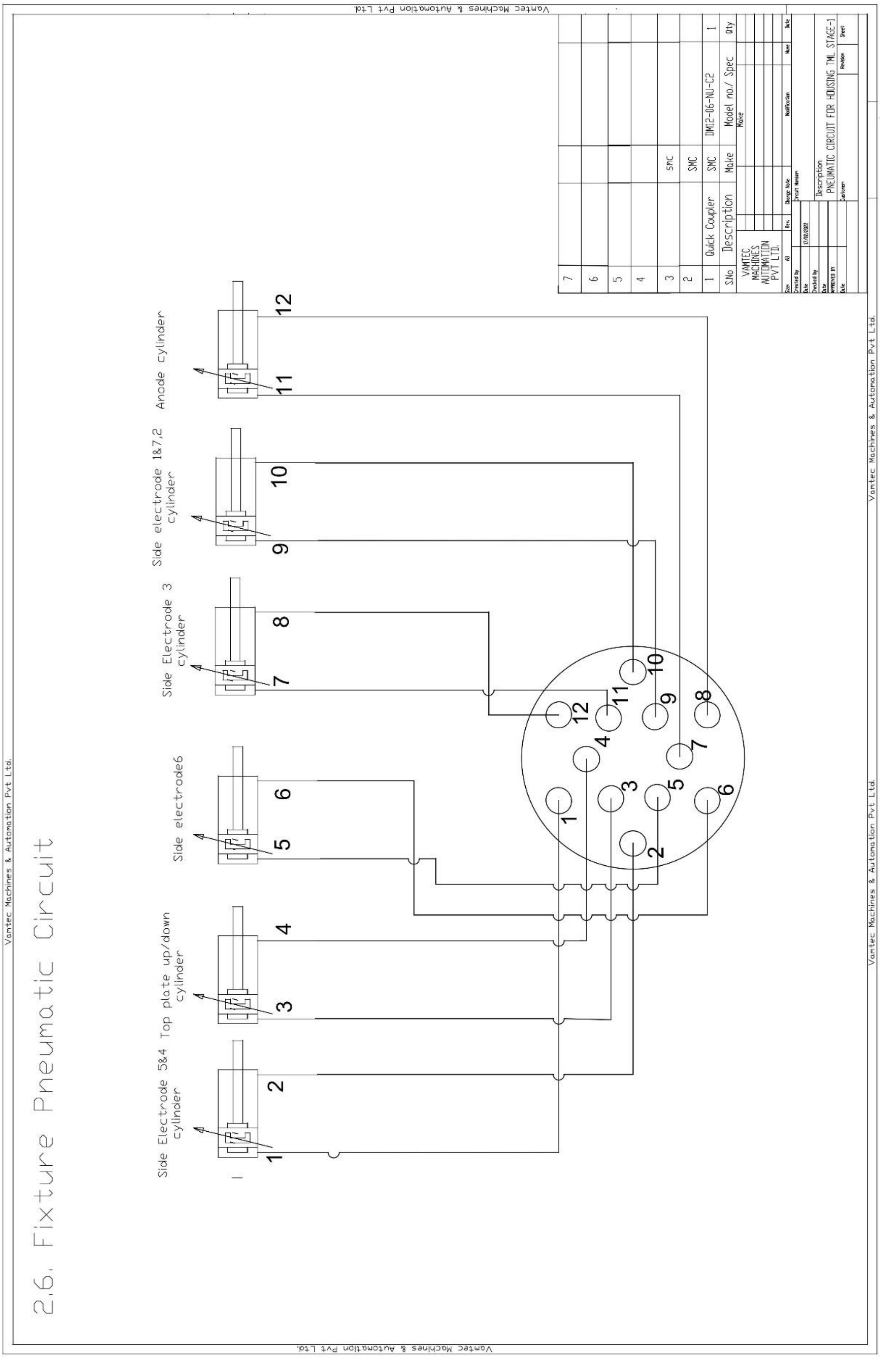


1	User	Date	TITLE: Fixture For 2.6 Front Plate
	Revised By	Date	
Drawn By	User	Date	VAMTEC Machines & Automation Pvt Ltd.
Checked By	User	Date	
	Signature	Date	

INSPECTION DIMENSION

DWG NO.	QUANTITY 1 No	MATERIAL:	FINISH:	UNLESS OTHERWISE SPECIFIED: SURFACE FINISH:None TOLERANCES: LINEAR: ±0.05 ANGULAR: ±1°
Third Angle Projection	DEBURR AND BREAK SHARP EDGES	A3	SCALE:1:2	SHEET 5 OF 5

8. PNEUMATIC CONNECTION DETAILS



9. WEAR PARTS LIST

S.NO	Part Name	Drawing number
1	Connector For Anode 2.2	VAM-P-459-01-18-00
2	PEEK Bush For Anode Guide Rod 2.6Fp	VAM-A-463-01-06-00
3	Anode Guide Bush For 2.2	VAM-P-459-01-13-00
4	Locating pin for 2.6 fixture	VAM-A-463-01-22 -00
5	Peek Bush in electrode mounting block	VAM-P-427-03-09 00
6	Side Electrode 1 & 7 Guide Plate For 2.6	VAM-A-463-01-25 -00
7	Changeable anode for 2.2	VAM-P-459-01-37-00